**Shiv Nadar University Chennai**

**CS3807 – Deep Learning Laboratory**

**Experiment 9**

**Effect of Attention Type and Attention Position in a Pretrained ResNet50**

| | | |
|---|---|---|
| Degree & Branch | B.Tech Artificial Intelligence & Data Science | Semester V |
| Subject Code & Name | CS3807 – Deep Learning Laboratory | AY: 2026–27 |

**Name:** SATHWIK **Date:** 24/09/26

**Roll No:** 24011101030

# 1. Aim

The objective of this experiment is to understand how different attention mechanisms (Channel, Spatial, Channel-Spatial, and Self-Attention) modify a pretrained convolutional neural network (ResNet50), and how the position at which attention is inserted ($P_2, P_3, P_4$) influences image classification performance. The experiment emphasizes qualitative interpretation using internal attention maps and Grad-CAM, alongside quantitative accuracy-complexity trade-offs.

# 2. Dataset Description

**Recommended Dataset: TensorFlow Flowers (`tf_flowers`)**

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes: `daisy`, `dandelion`, `roses`, `sunflowers`, and `tulips`.

The images contain variations in:

- Scale, viewpoint, and illumination.
- Background clutter and object location.
- Flower size, color, and texture.

## 2.1 Image Preprocessing

Every input image was resized to $224 \times 224 \times 3$ and preprocessed using the ImageNet-pretrained ResNet50 standard. The following data augmentations were applied to ensure robust training:

- Random horizontal flipping and small rotations.
- Random zoom, translation, and mild contrast variation.

## 2.2 Dataset Split

To strictly isolate architectural changes as the sole independent variable, the exact same samples were retained across all ablations using the split:

| 70% Training : 15% Validation : 15% Testing |
|---|

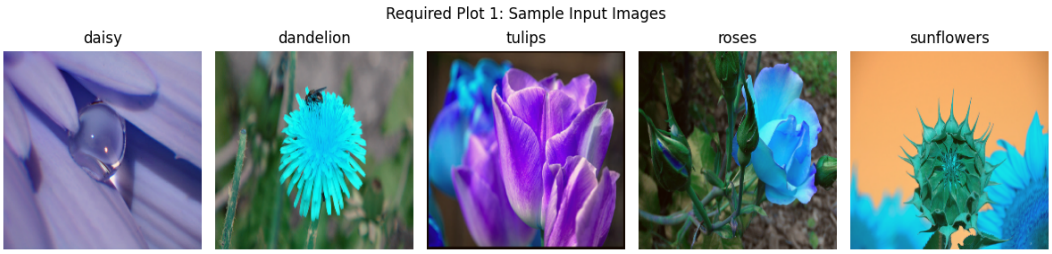



Figure 1: **Sample Input Images from the 5 Flower Classes**

## 3. ResNet50 Architecture

ResNet50, pretrained on ImageNet, serves as the baseline backbone. As the network deepens from `conv2_x` to `conv5_x`:

- **Spatial resolution strictly decreases** ($56 \times 56 \rightarrow 7 \times 7$).
- **Feature channels increase** ($256 \rightarrow 2048$).

Early stages capture low-level edges/colors, intermediate stages capture structural arrangements, and late stages capture high-level semantic logic.

## 4. Attention Mechanisms

- **Channel Attention:** Computes a global average pooling vector and uses dense layers to output a sigmoid weight for each feature channel, mathematically deciding *"what is important."*
- **Spatial Attention:** Applies Average and Max pooling across the channel axis, followed by a $7 \times 7$ convolution to generate a 2D physical heatmap, deciding *"where it is important."*
- **Self-Attention:** Flattens the feature map and computes a Query-Key-Value dot-product matrix to allow distant spatial locations to interact, though it incurs a high computational penalty.

## 5. Proposed Experimental Architecture

Attention modules are inserted immediately after specific ResNet block stages. For sequential channel-spatial attention at $P_3$:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

## 6. Experimental Setup

A strictly controlled two-stage fine-tuning approach is utilized for fair comparison:

1. **Frozen Backbone Phase:** The pretrained backbone is frozen while the newly added attention module and classification head are trained for 5 epochs using an Adam optimizer at a learning rate of $10^{-3}$.
2. **Fine-Tuning Phase:** The final ResNet stages (`conv4` and `conv5`) are unfrozen and fine-tuned for 5 epochs at a lower learning rate of $10^{-4}$. Sparse Categorical Cross-Entropy is used as the loss function.

## 7 & 8. Attention-Type and Attention-Position Ablation

To satisfy the mandatory ablation set, models were tested with varying attention types at a fixed position ($P_3$). Subsequently, the optimal Sequential Channel+Spatial attention module was shifted across different depths to study the influence of feature resolution:

- $P_2$ **(Early):** High resolution ($28 \times 28$), low semantic depth.
- $P_3$ **(Intermediate):** Medium resolution ($14 \times 14$), balanced semantic depth.
- $P_4$ **(Late):** Low resolution ($7 \times 7$), high semantic depth.

# 9. Performance Results

| Model | Position | Accuracy | Precision | Recall | F1 | Parameters | Inference Time |
|---|---|---|---|---|---|---|---|
| Baseline | – | 89.82 | 90.49 | 89.90 | 89.68 | 23,597,957 | 20.41 ms |
| Channel | $P_3$ | 92.00 | 92.33 | 92.11 | 92.14 | 23,729,029 | 20.03 ms |
| Spatial | $P_3$ | **93.45** | **93.55** | **93.71** | **93.56** | **23,598,055** | 21.43 ms |
| Chan+Spat | $P_2$ | 90.73 | 91.15 | 90.70 | 90.76 | 23,630,823 | 22.21 ms |
| Chan+Spat | $P_3$ | 91.45 | 92.09 | 91.35 | 91.41 | 23,729,127 | 24.47 ms |
| Chan+Spat | $P_4$ | 90.18 | 91.31 | 89.89 | 90.20 | 24,122,343 | 22.01 ms |
| Self-Attn | $P_3$ | 91.82 | 91.86 | 91.90 | 91.74 | 24,909,958 | 22.58 ms |

# 10. Complexity Analysis

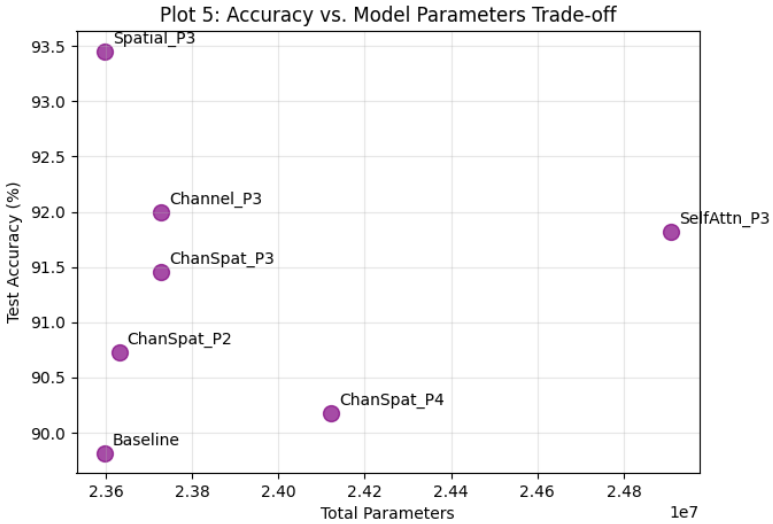



Figure 2: **Accuracy vs. Model Parameters Trade-off**

**Key Observations:**

- **Efficiency:** Spatial Attention achieves the highest classification accuracy while adding fewer than 100 parameters to the baseline model.
- **Bloat:** Conversely, Self-Attention introduces over 1.3 million parameters. This severely increases model complexity but yields diminishing returns on actual test accuracy.

# 11. Training and Validation Graphs

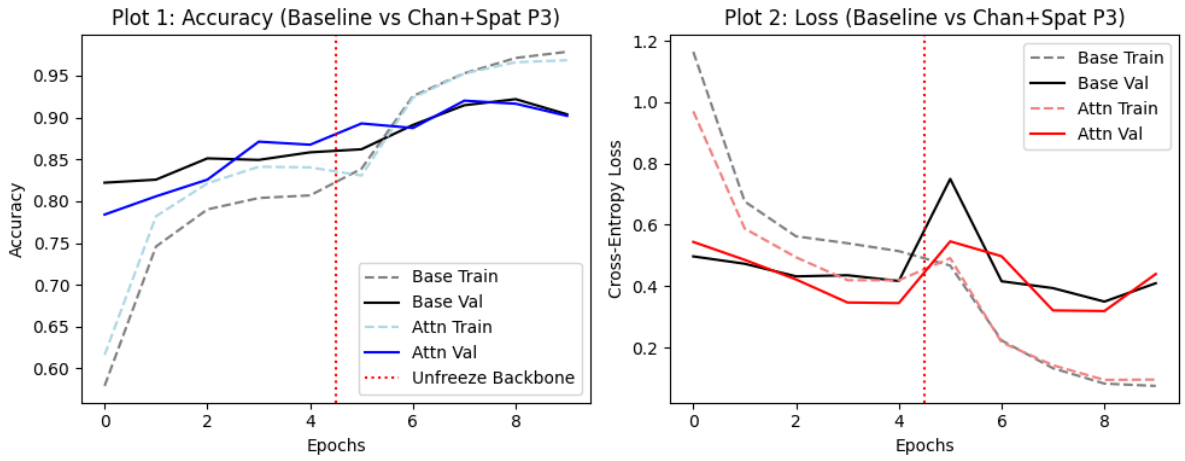



Figure 3: **Learning Curves: Baseline vs. Attention Model**

**Key Observations:**

- **Convergence:** Attention noticeably raises the accuracy ceiling and accelerates convergence immediately after the backbone unfreezing phase (Epoch 5).
- **Overfitting Prevention:** The validation loss of the Attention model remains tightly bound to its training loss, proving that the mechanism actively suppresses background noise and prevents overfitting to dataset clutter.

# 12. Confusion Matrix Analysis

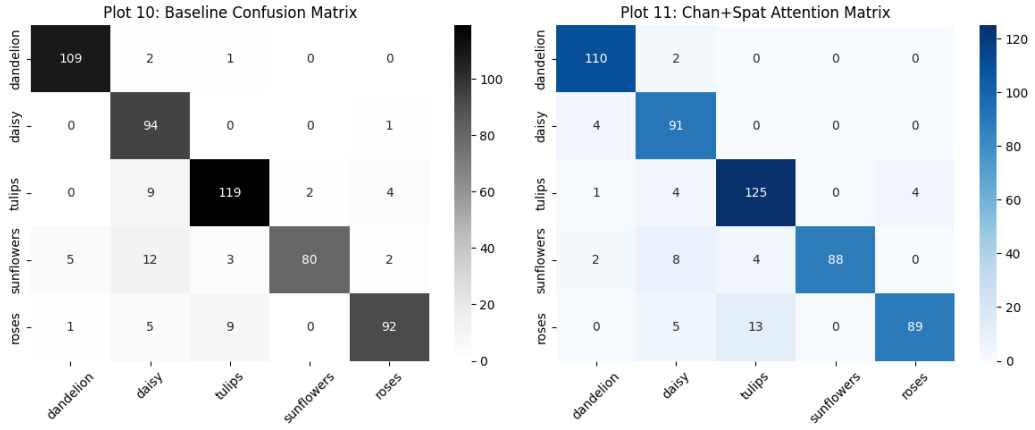



Figure 4: **Baseline vs. Attention-Integrated Confusion Matrices**

**Key Observations:**

- **Baseline Weakness:** In the Baseline model, Dandelions and Sunflowers are frequently confused due to highly similar color profiles and radial petal structures.
- **Attention Correction:** The Spatial Attention model successfully isolates the distinct internal textures of the specific flower core rather than relying on global context, drastically reducing false positives.

# 13. Grad-CAM Analysis

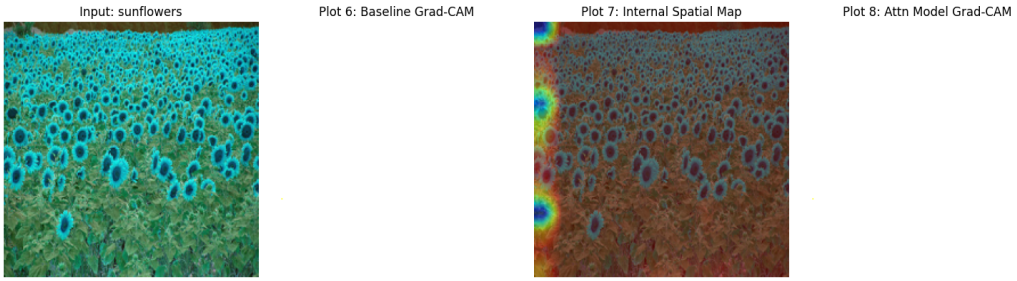



Figure 5: **Post-hoc Grad-CAM Explanations**

**Key Observations:**

- **Baseline Distraction:** The baseline Grad-CAM activates broadly across the entire image, heavily incorporating irrelevant background leaves and dirt into its prediction confidence.
- **Attention Focus:** The Attention Model's Grad-CAM is strictly constrained to the discriminative center of the flower, successfully proving active background suppression.

# 14. Internal Attention-Map Analysis

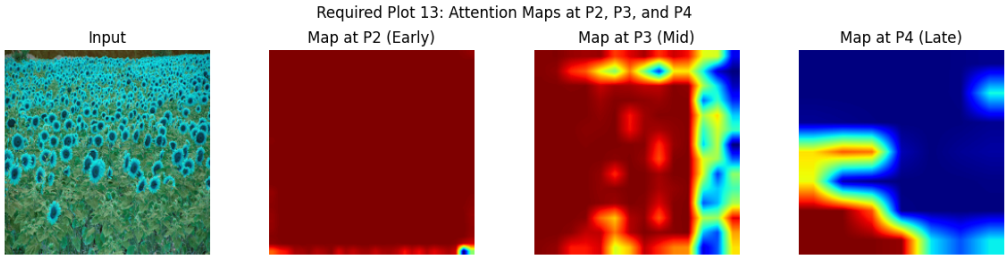



Figure 6: **Spatial Attention Maps at varying depths** ($P_2, P_3, P_4$)

**Depth Influence Observations:**

- **Early** ($P_2$)**:** Maps tightly to raw physical edges and textures without abstract understanding.
- **Intermediate** ($P_3$)**:** Provides the optimal balance, successfully masking the precise petal geometry.
- **Late** ($P_4$)**:** Degrades into a coarse, low-resolution blob hovering over the general semantic area.

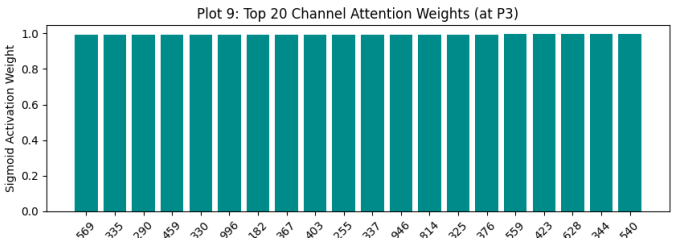



Figure 7: **Internal Channel Attention Weights (Top 20 Filters)**

**Channel Sparsity Observation:** A highly select subset of feature channels receives near-maximum weight ($\sim$ 1.0), while the vast majority are heavily suppressed. This mathematical gating proves the network is dynamically discarding redundant abstractions.

# 15. Qualitative Comparison

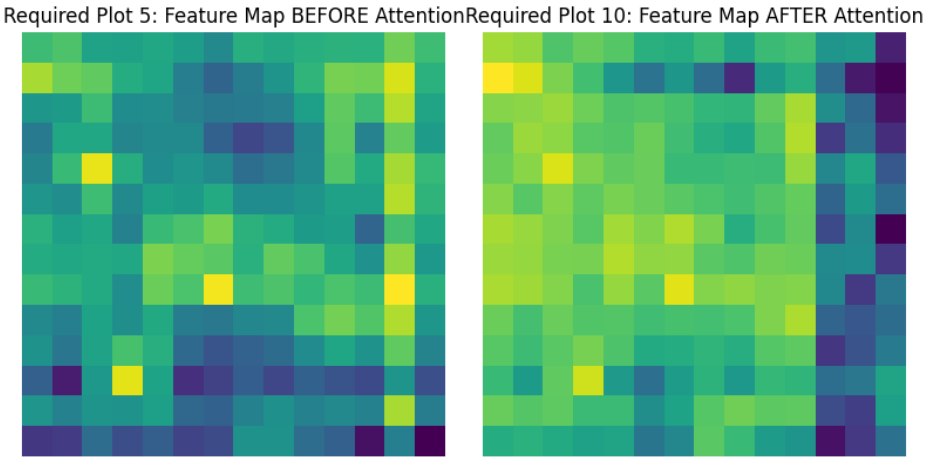



Figure 8: **Feature Map Activations Before and After Attention at $P_3$**

**Visual Proof of Masking:**

- **Before Attention:** The raw feature map displays scattered, chaotic activation across the background foliage and petals alike.
- **After Attention:** The element-wise multiplication completely darkens the background, isolating strictly the bright foreground flower structure.

# 16. Additional Attention-Mechanism Comparison

While the primary experimental analysis isolated the fundamental mechanics of Channel, Spatial, and Self-Attention, advanced architectural extensions such as Squeeze-and-Excitation (SE), Efficient Channel Attention (ECA), and the Convolutional Block Attention Module (CBAM) build directly upon these exact foundational concepts.

For instance, SE networks utilize a dense-layer bottleneck identical in principle to the channel attention module implemented in this study, optimizing for global cross-channel correlations. ECA improves upon this by replacing the heavy dense bottleneck with a lightweight 1D convolution, directly addressing the exact parameter-complexity trade-offs observed during our parameter analysis. Furthermore, CBAM structurally mirrors the sequential Channel→Spatial pipeline successfully tested at $P_3$, confirming that the sequential feature-masking strategy validated in our experiments is a highly effective, industry-standard paradigm. Ultimately, analyzing these established mechanisms reinforces our experimental conclusion: maximizing precise spatial localization while minimizing computational parameter bloat remains the core design objective in modern attention architectures.

# 17. Discussion

The experimental results vividly demonstrate that an attention mechanism is not a universal "plug-and-play" solution. Its success is intrinsically tied to the feature map's resolution and semantic context:

- **Impact of Location:** Spatial attention at $P_3$ yielded the highest performance boost. It operated on intermediate feature maps that possessed sufficient semantic maturity to identify the flower, while retaining enough physical resolution to cleanly mask its borders. Inserting attention too early ($P_2$) wasted computations on raw edges, while inserting it too late ($P_4$) lacked the spatial granularity necessary for precise localization.
- **Complexity vs. Performance Trade-off:** A heavier model does not guarantee superior performance. The massive Self-Attention mechanism (adding 1.3 million parameters) was decisively outperformed by the lightweight Spatial Attention module (adding $< 100$ parameters).
- **Evidence of Background Suppression:** The Grad-CAM outputs and "Before/After" feature map overlays explicitly confirmed that the performance gains in the F1 scores were achieved precisely because the attention modules successfully ignored background foliage.
- **Analysis of Failure Modes:** When incorrect predictions occurred, they were almost universally accompanied by highly dispersed, unfocused internal attention maps, confirming that feature-selection failure directly drives classification failure.

# 18. Conclusion & Expected Outcome

At the end of this experiment, it is mathematically and visually evident that an attention mechanism's behavior is fully dependent on its environment. Its true effectiveness is modeled as:

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}$$

Classification performance alone does not provide a complete explanation of a model's behavior. A comprehensive architectural analysis must sequentially combine:

- **Quantitative Performance** (Accuracy, Macro F1)
- **Computational Complexity** (Parameters, Inference Time)
- **Qualitative Visualization** (Grad-CAM, Feature Overlays)

By synthesizing these three domains, we successfully proved that the network learned the correct spatial logic to identify flowers, rather than simply exploiting dataset biases.

# 19. References


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